Music Generation with AI

An LSTM learns note patterns from MIDI files and composes new music, written out as a MIDI file and rendered to audio.

Prepared by · Sunny Gouria

Domain · Artificial Intelligence

Status · completed

CLAIM

The model reads a run of notes and predicts the next one, again and again, until a melody exists.

A TensorFlow-free Markov fallback produces music even without the deep-learning install.

Four stages from raw MIDI to new music

01 **Collect MIDI data**
A starter corpus, or your own .mid files.

02 **Preprocess**
music21 flattens each file into a note sequence.

03 **Train the LSTM**
A two-layer network learns to predict the next note.

04 **Generate and save**
Sampled notes become a MIDI file, then audio.

WHAT IT MEANS

It always produces music

A second-order Markov chain needs only music21 and numpy, so the project runs even without TensorFlow installed.

Built with · Python · TensorFlow

Source: model.py

How a piece is generated

01

Parse

MIDI read into notes and chords by music21.

02

Sequence

Fixed windows of the past twenty-four notes.

03

Predict

The LSTM samples the next note — the core step.

04

Write

Tokens rendered to MIDI, then to audio.

Temperature controls how adventurous the sampling is — below 1.0 stays close to the training data.

Source: model.py

Where the numbers stand on the sample run

60

MIDI FILES

Generated as the starter corpus.

4,328

NOTES PARSED

Across every file in the corpus.

156

VOCABULARY

Distinct notes and chords learned.

4,304

TRAINING WINDOWS

Sequence length 24, stride 1.

Source: sample corpus, run 16 Oct 2026

Four parts, one per module

Data pipeline

music21 parses MIDI into integer note sequences.

LSTM model

Two layers, 128 units each, dropout 0.3.

Fallback

A second-order Markov chain that needs no TensorFlow.

Output

MIDI saved, WAV rendered when fluidsynth exists.

Source: project structure

Train it on your own music

Drop .mid files into data/midi/, then run python main.py all. Full commands are in the README.

TASK 03 · COMPLETE

Built by · Sunny Gouria

Stack · Python · TensorFlow

Repo · CodeAlpha_MusicGen